

Can User Reviews Indicate Usability Heuristic Issues?

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ABSTRACT

User reviews of mobile applications in online stores have been used as a source for numerous studies. Many focus on indicating new functional requirements and extracting problems regarding quality characteristics, such as Usability. One of the most known strategies to evaluate Usability is Nielsen's heuristics. We believe that user reviews in online stores can indicate heuristics issues, thus supporting evaluations more focused on the user feedback. Therefore, this study aims to investigate the presence of usability heuristic issues through user reviews. To achieve this goal, we collected and analyzed 200 reviews, split into 528 statements, from 10 free and paid Android and iOS apps. Our results showed that user reviews have certain potential to indicate heuristic issues, suggesting new and challenging future research for the CHI community.

CCS CONCEPTS

• **Human-centered computing** → HCI design and evaluation methods; *Collaborative and social computing*; • **Software and its engineering** → Software usability.

KEYWORDS

usability heuristics, user reviews, online stores

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1 INTRODUCTION

Most mobile applications (or apps) are obtained through online stores nowadays, such as *Google Play Store* or *Apple App Store*. These stores allow users to report their opinion on their use, which we call here as *User Reviews*. According to Hedegaard and Simonsen [7], reviews are “a piece of text detailing pros and cons of a product and possibly an assessment of it and recommendations for potential buyers, written by a user of the product who has been in possession of said product and used it for some time”.

Popular apps like Facebook usually receive around 427 reviews per day [16]. This feedback is essential for practitioners to improve their applications' quality by adding new features, improving existing ones, or fixing bugs [17]. Additionally, these reviews have been a source for several studies, such as requirements elicitation [9], usability evaluation [7] and users' intentions extraction [17].

A study by Groen et al. [6] found out that these reviews can address quality characteristics by which users are directly affected, such as Reliability and Security, and its subcharacteristics. The authors stated that the main quality attribute with criticisms in these reviews is Usability, which is a characteristic related to the ease of learning and user satisfaction as a result of the usage of an interactive system [14].

Jakob Nielsen proposed ten general principles for Usability called “heuristics” [14]. They are desirable characteristics of interaction and interface and can be used for usability evaluations. They are listed as follows: H1. Visibility of system status, H2. Match between system and the real world, H3. User control and freedom, H4. Consistency and standards, H5. Error prevention, H6. Recognition rather than recall, H7. Flexibility and efficiency of use, H8. Aesthetic and minimalist design, H9. Help users recognize, diagnose and recover from errors and H10. Help and documentation.

These heuristics are often used in inspections, allowing experts to understand usability issues in the user interface and interaction [15]. However, we wonder if reviews written by users may indicate heuristic-related issues. If this is the case, user reviews could help prioritize efforts in future HCI evaluations.

Moreover, we often see studies regarding reviews using basic dimensions of Usability to categorize user problems [6, 12]. However, by using Nielsen's heuristics, it is possible to categorize problems into more groups, representing a higher level of granularity, and can help practitioners more precisely. Such categories (i.e., the ten heuristics) are commonly known and used to reason usability issues. Therefore, our research question is “*To what extent do user reviews indicate problems related to usability heuristics?*”. Thus, this paper aims to investigate the identification of evidence of Usability Heuristics from the user reviews in online stores. We expect to provide a new perspective for future work to improve usability problems' identification through reviews.

2 RELATED WORK

There are several studies using reviews as a source for extracting meaningful information. Groen et al. [6] aimed to understand how feedback from the online stores could be helpful to assess quality characteristics and if there is potential to automate it. They found out that users mainly write about Usability and Reliability. They also defined a set of 16 language patterns for Usability and verified that they correctly identified 1,528 statements from a large

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dataset, showing good automation potential. This work inspired us to investigate reviews from the perspective of usability heuristics.

Lu and Liang [11] suggest that Reliability, Usability, Portability, and Performance are indeed indicated in the user reviews, but their focus was not to show to what extent each one is addressed. Instead, their focus was to compare classification techniques on how well they classify user reviews.

Hedegaard and Simonsen [7] analyzed 3,492 reviews to investigate the presence of Usability and User Experience dimensions, such as Memorability and Learnability, Efficiency, among others. This study is similar to ours in analyzing reviews according to specific aspects.

There are also textual evaluation methods that take user reviews as input for evaluating the HCI of an interactive system since they report interaction experiences, whether long-term or initial [13]. For instance, the MALTU methodology [13] classifies user posts extracted manually or automatically from a social system to evaluate Usability and User Experience aspects. Such methodology has already been applied in the evaluation of some systems' domains, such as healthcare systems [4], educational systems [13], mobile applications [3], and ubiquitous systems [5].

Therefore, these studies portray the possibility of analyzing Usability through user reviews and analyzing and finding patterns for more specific Usability levels. However, we have not seen studies that perform this type of analysis from the perspective of Usability Heuristics. Thus, our focus is to explore this insightful idea.

3 METHODOLOGY

To achieve our goal, we applied a methodology consisting of four sequential steps based on the procedures performed by Groen et al. [6]. In the following subsections, we describe each step.

3.1 Step 1: Selecting and Obtaining Reviews

First, we established the mobile applications from which user reviews were selected. To ensure a representative dataset of user reviews, we defined that the application should (i) belong to one of the five categories identified by Pagano and Maalej [16] with the highest number of user reviews: entertainment, social networks, communication, productivity, and games; (ii) have more than two reviews for each rating on the five-star scale, so it is possible to obtain a sample that contains two reviews rated with each star; and (iii) be available on two online stores: *Google Play Store*¹ and *Apple App Store*². In each category, we selected a paid app and a free app. We defined 20 data sources from the selection criteria: five application categories * two applications per category (one paid and one free) * two online stores. We selected the applications during January and February 2021 from the USA online stores, which provide user reviews in English.

After this selection, we developed 2 *scripts* (one for each online store) to obtain user reviews using the *Python* programming language and the libraries *Google Play Scraper*³ and *App Store Scraper*⁴. Each *script* was performed for each of the ten apps, producing 20

data sources. The *scripts* obtained for each review their textual content and star rating, the application name, and the associated online store. All reviews were unified in one dataset and exported as a comma-separated values (csv) file. This dataset contains 77,665 user reviews. Table 1 summarizes the results of this step.

Table 1: Selected Applications and Their Amount of Reviews

Application	Category	Monetization	Reviews obtained
Netflix	Entertainment	Free	10,000
Akinator	Entertainment	Paid	8,845
Google Docs	Productivity	Free	10,000
eDrawings	Productivity	Paid	591
Facebook	Social Networks	Free	7,600
TweetCaster Pro	Social Networks	Paid	10,000
Telegram	Communication	Free	10,000
TeamSpeak	Communication	Paid	3,497
Brawl Stars	Games	Free	10,000
NBA2K20	Games	Paid	7,132

3.2 Step 2: Preparing Reviews

This step aimed to prepare the set of reviews to be classified. This preparation is necessary because user reviews have an unstructured format. Then, we first filtered the dataset collected in Step 1 by removing reviews with (i) blank text content, (ii) emojis-only content, and (iii) less than two words. This filtering process excludes 8,588 reviews, remaining 69,077 user reviews in the dataset.

After that, we randomly selected two user reviews for each of the five stars from the 20 data sources, totaling 200 reviews in our dataset. Then, we decided to split the reviews into their statements so that the researchers could classify each statement in a review. The reason is to make the classification process more specific. We developed a *script* using the *Python* programming language to perform this task. This *script* splits the user reviews into statements by "." and by "(line break)", resulting 530 statements. We discarded two reviews not written in English, remaining 528 statements. Finally, these statements were exported in csv format to facilitate data storage and integration with other tools.

3.3 Step 3: Classifying and Consolidating Statements

Three researchers performed the classification: one undergraduate student, one M.Sc. student, and one Ph.D. professor, all working in the HCI field. Each read a given statement and assigned one or more heuristics representing the statement. If the researcher judges that no heuristics represent the statement, it is classified with no heuristics. Also, the researcher informed his/her confidence level for that classification and how much he/she thinks that statement corresponds to the classified heuristic. The researchers performed the classification in a *Google* spreadsheet created individually for each researcher to avoid bias during classification.

The confidence and correspondence level was defined according to the seven-point Likert scale. The confidence level scale has the

¹<https://play.google.com/store>

²<https://www.apple.com/br/app-store/>

³<https://pypi.org/project/google-play-scraper/>

⁴<https://pypi.org/project/app-store-scraper/>

following values: from (1) Not at all confident to (7) Completely confident. The correspondence level scale has the following values: from (1) Does not match at all to (7) Matches fully.

To analyze the agreement between the three researchers' classifications for the 528 statements, we calculated the percentage agreement and performed Cohen's kappa [2] for each pairwise combination of two researchers, then calculated the average kappa. We interpreted the kappa results as suggested by Landis and Koch [10]. First, regarding whether or not there are usability heuristics-related issues, the researchers fully agreed on 366 statements (.69 percentage agreement). We observed a fair agreement between the researchers' classifications: $\kappa = .321$ (95% confidence interval, .224 to .418), $p < .001$. Second, regarding the statements classification on one or more usability heuristics, the researchers fully agreed on 344 statements (.65 percent agreement). In this case, we observed a slight agreement between the researchers' classifications: $\kappa = .194$ (95% confidence interval, .132 to .256), $p < .001$.

After the individual classification, discussion sessions took place with all researchers to achieve a common classification of statements classified differently by one or more researchers. Also, researchers discussed classifications with confidence levels lower than five, totaling 251 statements for this consolidation step. After reaching a consensus, the classification data was updated.

3.4 Step 4: Analyzing Data

This step aimed to perform a quantitative analysis of classifications using descriptive statistics. We mainly analyzed the proportion of found heuristics, their co-occurrence and their relation to (i) statement length, (ii) the way the application is monetized, (iii) the category of applications, and (iv) the online store.

4 RESULTS

As a general result, we observed that from 528 statements, 132 were classified with Nielsen's usability heuristics (25% of the statements). Thus, the remaining 396 statements were classified without heuristics (75% of the statements). In the following subsections, we present more specific results from our analysis.

4.1 Occurrence of Usability Heuristics

Figure 1 shows the number of issues for each usability heuristic found in the statements. While H9 (help users recognize, diagnose, and recover from errors) was the heuristic that occurs the most (46 times), H5 (error prevention) was the heuristic that occurs the least (2 times). Table 2 presents a sample of statements classification.

4.2 Statement Length and the Occurrence of Heuristics

We hypothesized that statements with more text could be more likely to indicate issues, given the presence of more detail. Therefore, we decided to analyze statements length and the occurrence of heuristics. We observed that the average length of the statements in our dataset is 15 words, and the standard deviation (SD) is 12 words. Moreover, we identified that the average length of statements classified with no heuristics is 13 words (SD = 11). In comparison, the average length of statements classified with heuristics is 21 words (SD = 15).

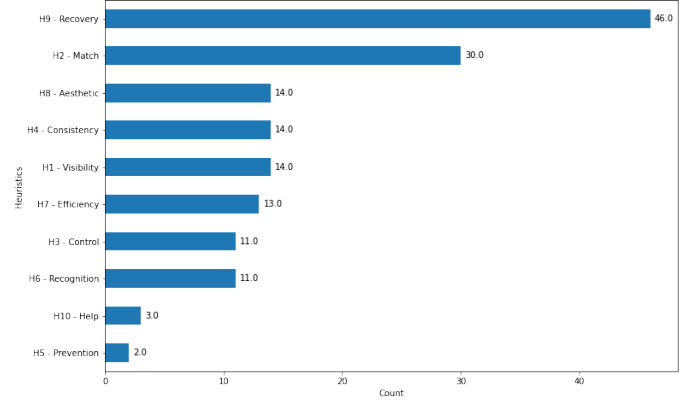


Figure 1: Proportion of heuristic issues in the statements

Table 2: Examples of Statements Classification

Statement	Heuristic
"I'm happy to help improve it, but i don't understand why it's so easy to destroy your format, and so hard to make it look presentable."	H1
"And this game won't allow me to get past creating my character."	H3
"Moving the icon that allows you to go forward or backwards in the video doesn't fix the issue, as a matter of fact if you try to move the icon before the buffering starts the video freezes at the exact same spot."	H9

Regarding the applications' monetization, the average length of statements classified with heuristics in paid applications is 18 words (SD = 13). In contrast, the average length of statements classified with heuristics in free applications is 24 words (SD = 16 words).

4.3 Applications' Monetization and the Occurrence of Heuristics

Figure 2 shows the proportion of usability heuristic issues in paid and free applications. Paid applications (*Akinator*, *eDrawings*, *TweetCaster Pro*, *TeamSpeak*, and *NBA2K20*) has 246 statements. 61 (25%) are classified with heuristics, and 185 (75%) are classified with no heuristics. While H9 (help users recognize, diagnose, and recover from errors) is the most recurrent heuristic (27 times) in paid applications, H4 (consistency and standards) appears only once. H5 (error prevention) does not appear in the statements of paid applications.

Free apps (*Netflix*, *Google Docs*, *Facebook*, *Telegram*, and *Brawl Stars*) has 282 statements. 71 (25%) were classified with heuristics, and 211 (75%) were classified with no heuristics. While H2 (match between system and the real world) is the most recurrent heuristic (20 times) in free applications, H10 (help and documentation) appears only once.

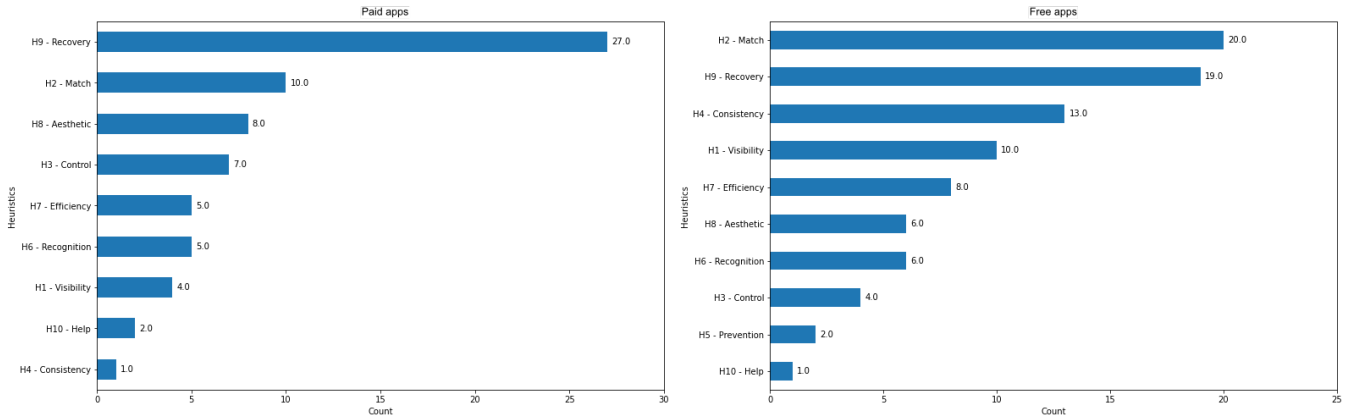


Figure 2: Proportion of usability heuristic issues in (a) paid apps and (b) free apps

4.4 Application Categories and the Occurrence of Heuristics

We also would like to understand if there was a difference of heuristics occurrence regarding categories of apps. *Games* is the category with most statements (125), followed by Productivity (106), Social Networks (105), Entertainment (102), and Communication (90). Figure 3 presents an overview of the occurrence of heuristics for each category. Analyzing specifically the heuristics present in the statements of each category, we observed that the statements in Games contain six heuristics, the smallest number of heuristics among the application categories.

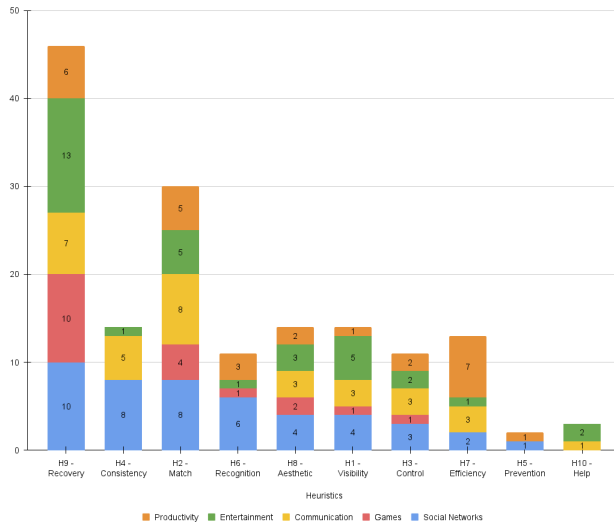


Figure 3: Proportion of usability heuristic issues for each application category

4.5 Online Stores and the Occurrence of Heuristics

In the *Google Play Store*, we have 36 statements classified with heuristics out of 155. These classified statements have an average

length of 18 words (SD = 16). While in the *Apple App Store*, we have 96 statements classified out of 373. The average length is 22 words (SD = 15). Figure 4 presents the occurrence of heuristics on both stores. While H9 (help users recognize, diagnose, and recover from errors) appears the most (12) in the *Google Play Store* statements, H10 (help and documentation) appears only once, and H5 (error prevention) did not appear in these statements. Similarly, H9 is the most recurrent heuristic (34) in the *Apple App Store* statements, whereas H10 appears only twice.

4.6 Co-occurrence of Heuristics

There are 24 statements classified with more than one heuristic. Therefore, we decided to understand and explore whether there could be a pattern of heuristics occurring at the same time. Figure 5 presents a co-occurrence graph with heuristics as nodes and the co-occurrence frequency as the weights between nodes. H2 (match between system and the real world) is the heuristic that most co-occurs (18 times), whereas H10 (help and documentation) is the heuristic that least co-occurs (two times). H5 (error prevention) did not co-occur with any other heuristic.

The largest match in the entire dataset is H2 (match between system and the real world) and H3 (user control and freedom), occurring six times. The second one is H2 and H4 (consistency and standards), occurring four times. The heuristic H2 occurs with seven other heuristics, except for the heuristic H10 (help and documentation). Additionally, we observed that the common co-occurrence involves only two types of heuristics. However, despite being rare, there can also be a co-occurrence between three types of heuristics. We observed this scenario twice, involving the heuristics H1 (visibility of system status), H2 (match between system and the real world), H3 (user control and freedom), H6 (recognition rather than recall), and H8 (aesthetic and minimalist design).

5 DISCUSSION

User reviews are a source for extracting several kinds of meaningful information, such as bugs, user needs, functional and non-functional requirements (NFRs). Regarding NFRs, reviews can address not only Usability, but characteristics such as Accessibility [1],

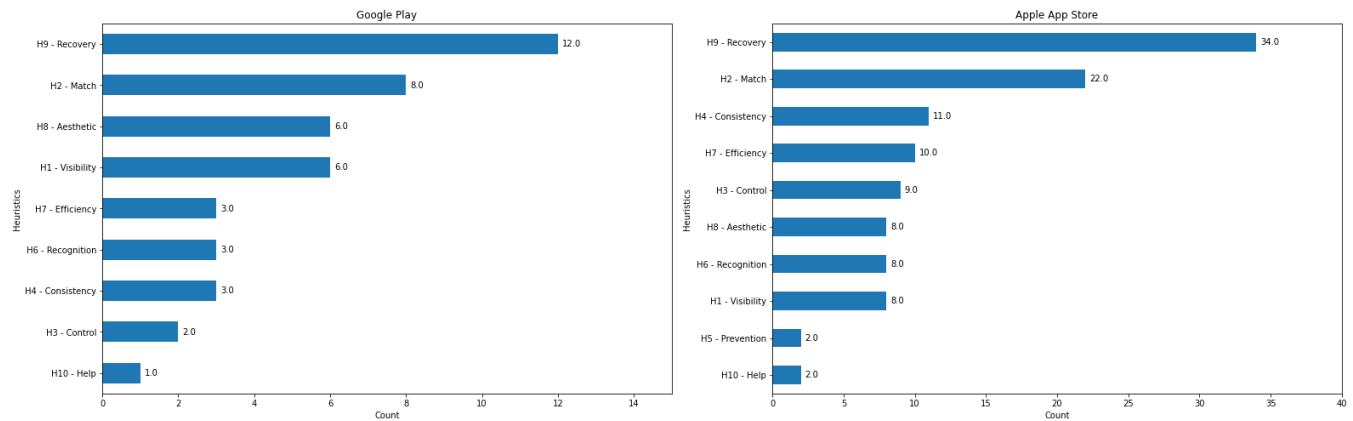


Figure 4: Proportion of usability heuristic issues in (a) Google Play Store and (b) Apple App Store

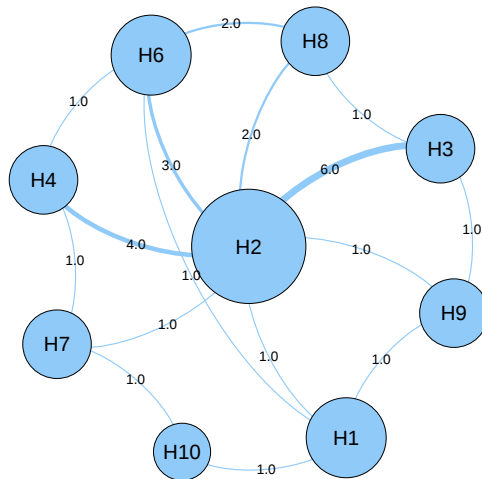


Figure 5: Co-occurrence graph between heuristics

Security and Reliability [6]. Regarding Usability, our results revealed that despite the small number of classified statements (25%), user reviews have a certain potential to indicate Nielsen's usability heuristic issues. The most occurred heuristics were H9 (help users recognize, diagnose and recover from errors) and H2 (match between system and the real world). Thus, users report mainly defects, failures, difficulties in understanding how the application works according to their real world vision. However, the one with the lowest occurrence was H5 (error prevention), which deals with interaction design strategies to prevent the user from making errors. Identifying this heuristic in statements was challenging because such issues would be simple to identify when analyzing application interfaces, but users do not elaborate on this kind of information in their reviews. This is a limitation compared to the use of heuristics in inspections: it is not possible to determine the place of occurrence for all reported issues, as it depends on the spontaneity of users when writing their reviews.

Games was the application category with the least number of different heuristics identified. This may be related to the fact that

in most statements, users reported dissatisfaction with the game's business rules instead of describing problems that impacted their use, such as: "*The first reason is that the chances of getting a new brawler is NEARLY impossible like the power points and coins are easy to get but somehow, the Big box and the mega box percentages are just the same.*" In these cases, we chose to classify the statement with no heuristics.

Overall, this manual classification is challenging because users often do not elaborate on the context of use so that the implied information can be clearly understood. Thus, approaches that use semantic and/or sentiment analysis (e.g., SentiStrength-SE [8]) can bring advantages to heuristics identification in future research.

We were able to identify potential, even though small, for automation. For instance, for H1 (visibility of system status), expressions like “*I don’t understand...*”, “*I can’t see...*”, “*where is it...*” usually appear, expressing the idea of a lack of feedback or understanding. This possible automation must first consider the statements’ length. As reported in the results, statements with usability heuristic issues usually have a higher number of words (21 in average), indicating that the user frequently details more information when reporting, although not knowing, heuristic issue. Automated tools can be built to enable this classification task from the identification of language patterns in user reviews, as performed in the study of Groen et al. [6] or also by employing machine learning techniques.

Based on our findings, we consider that textual evaluation based on usability heuristics could be used as a support technique for the usability evaluation process rather than an independent method. Also, applying this strategy can help identify the most critical portions of the system from the perspective of real users. These portions could be later evaluated with other methods, such as inspection methods (e.g., Heuristic Evaluation [15]), and those that directly involve user participation (e.g., user testing and user diary).

As the main contribution, our results reveal evidence that reviews can be used as a source of information on usability heuristic issues. We highlight challenges found in this identification process as well as opportunities for automated approaches and their use as a support technique for usability evaluation. The scripts and datasets constructed in this study are available as supplemental material attached to this paper, which can be used in further research.

6 FINAL CONSIDERATIONS

Through this study, we observed that user reviews have a certain potential to indicate usability problems through Nielsen's usability heuristics. However, classifying user reviews manually requires technical knowledge and a lot of effort on the part of classifiers. We are currently defining a set of language patterns as an automatic classification approach from the classification discussed in this paper. Also, we intend to investigate machine learning techniques as a way to automatically classify the user reviews based on Nielsen's usability heuristics.

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